

Coffee Shop Sales Analysis

Mai Nguyen Xuan Kieu

kieumnx@gmail.com



Table of Contents

01 About Dataset

02 Processing

03 Analysis

04 Summary

05 Proposal



About Dataset

About Dataset

1. Source

Kaggle: Coffee Shop Sales Analysis

2. Data content

- This dataset contains the sales data of a coffee shop
- Period: Jan 01, 2023 - Jun 30, 2023

3. Num of Columns: 18

transaction id, transaction date, transaction time, store id, store location, product id, transaction quantity, unit price, total bill, product category and many others







Processing

Tools

python™

Microsoft® SQL Server®

Power BI



Processing Steps

1. Check quality data and change data type

- Change data type of column transaction_date
- Remove duplicates

2. Create data model

- Split dataset into multiple tables: orders and order_items
- Create a new table: dim_date

3. Add column category_type into order_items in 4 groups

- Ingredients
- Finished Products
- Complementary Products
- Branded

4. Replace values Size for some product_id

5. Find bundled products

- Define a function BundledProducts()
- Using combinations function in Python
- Finding TOP 3 bundled products or something like that







Analysis



Overview

In the first 6 months of the year

Total Revenue is

$698.81 K

With

116.79 K

Total Orders

AND

$214 K

Total Sales

13.56 %

AVG Revenue Growth Rate

$6.38

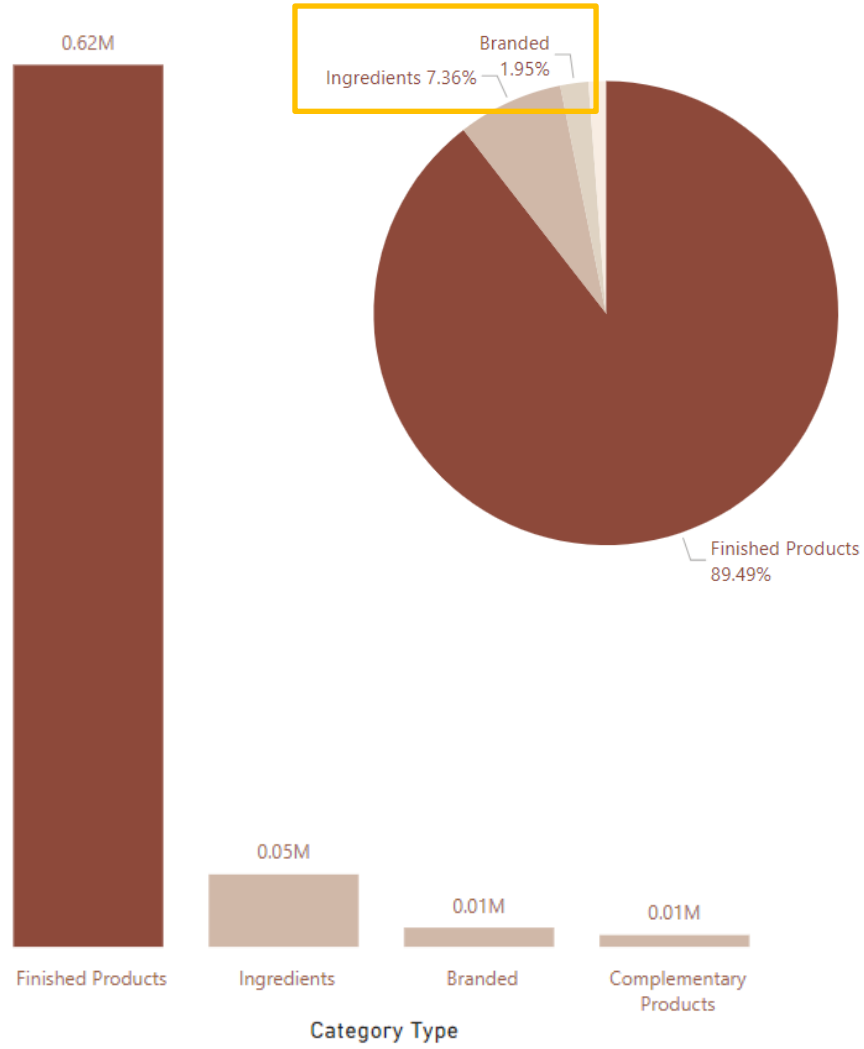
Average Order Values

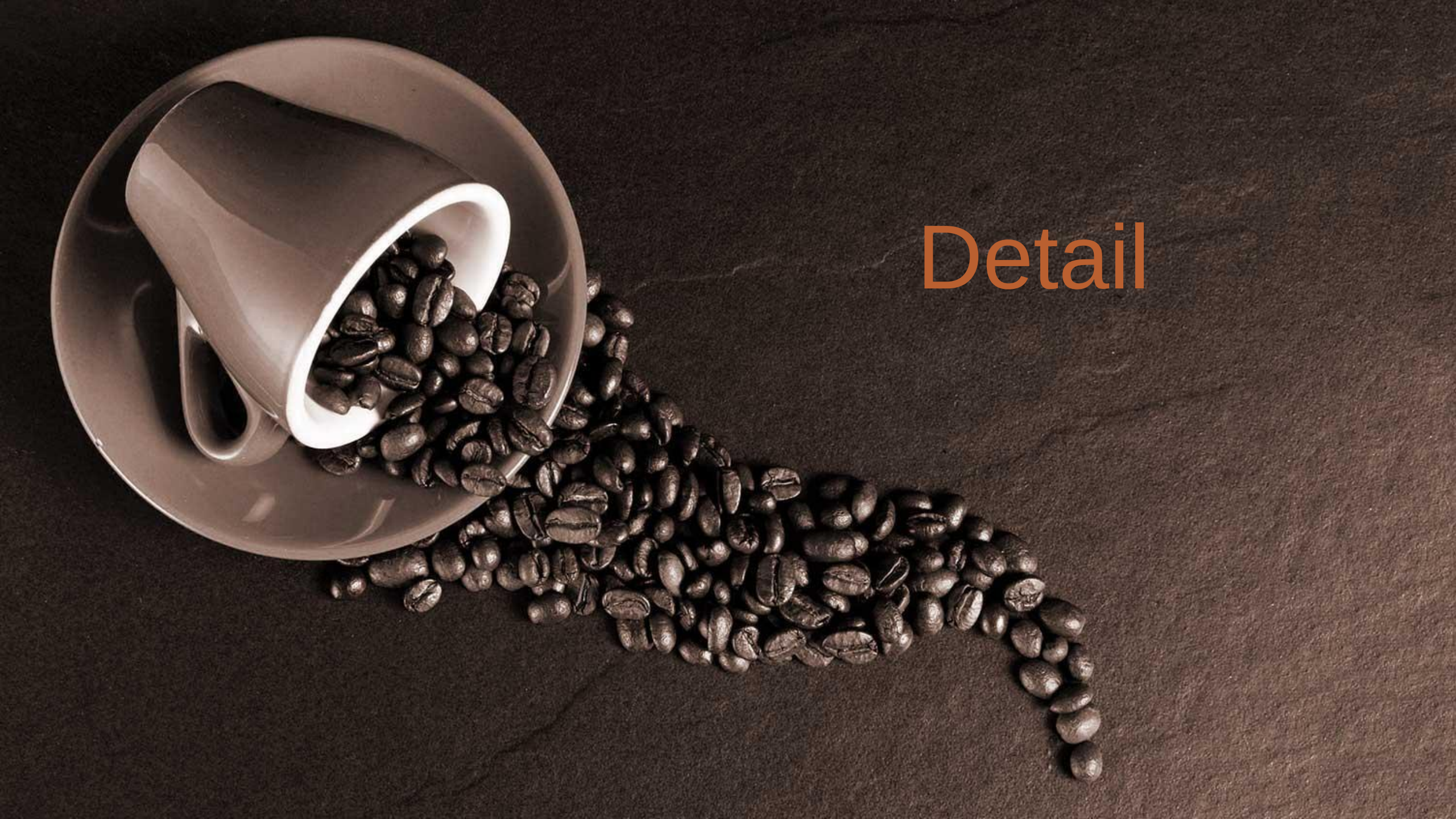
Finished Products has highest revenue with more than **$0.62 million**

Product Category	Product Type
Bakery	Biscotti
Bakery	Pastry
Bakery	Scone
Coffee	Barista Espresso
Coffee	Drip coffee
Coffee	Gourmet brewed coffee
Coffee	Organic brewed coffee
Coffee	Premium brewed coffee
Drinking Chocolate	Hot chocolate
Packaged Chocolate	Drinking Chocolate
Packaged Chocolate	Organic Chocolate
Tea	Brewed Black tea
Tea	Brewed Chai tea
Tea	Brewed Green tea
Tea	Brewed herbal tea

Other category types have pretty low revenue, just take **10%** of grand total

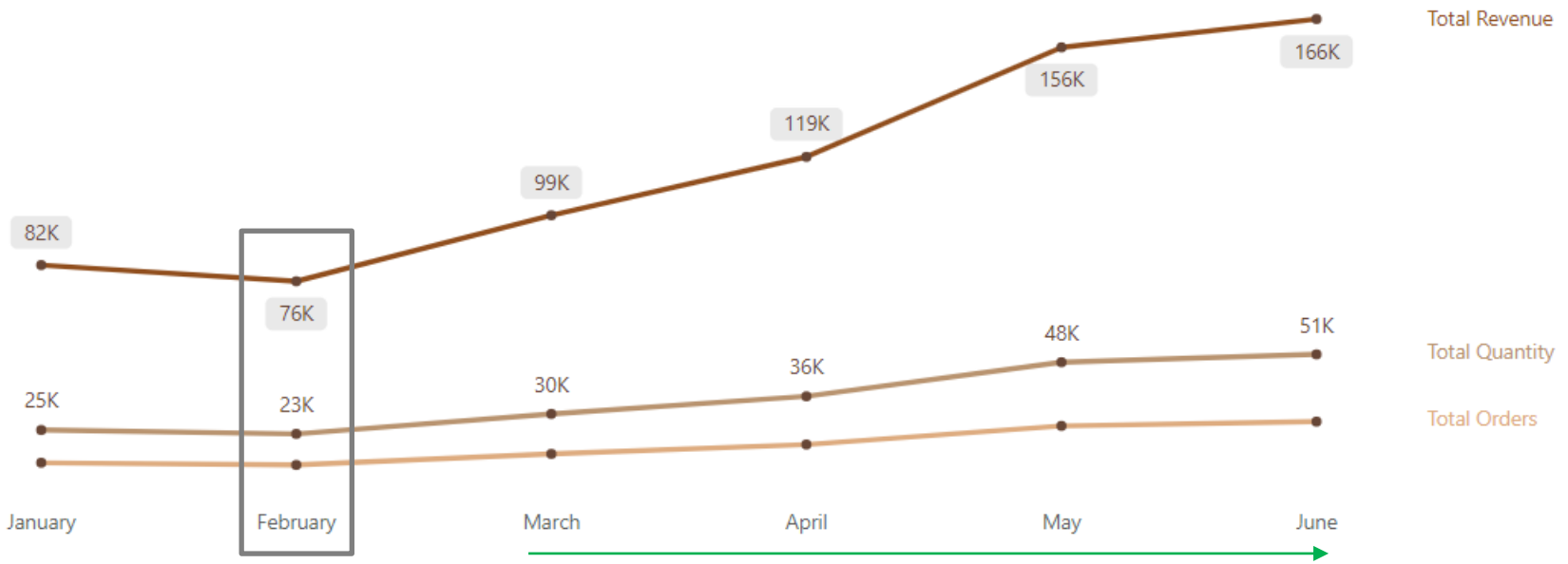
Total Revenue By Category Type




Detail

Time

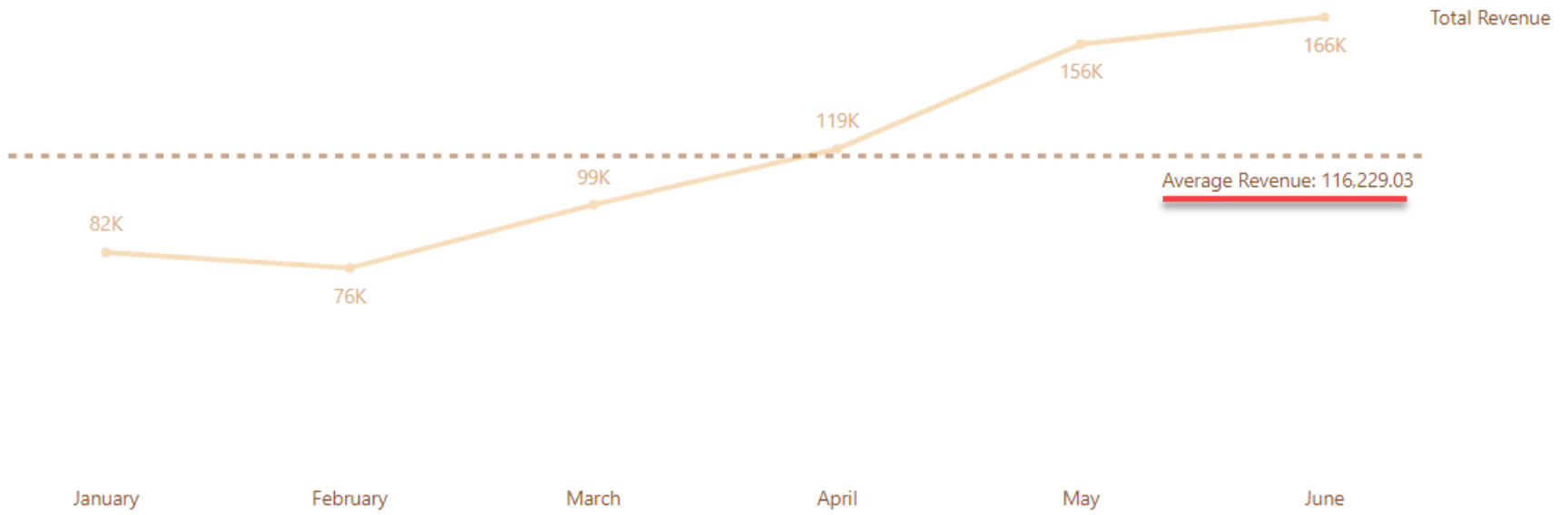
Total revenue, total quantity and total orders are tend to increase in the first half 2023



- Everything are tend to **decreased** in **February** and gradually increased in the following months
- **Revenue increased** quite rapidly from **March to June**
- On the contrary, sales and number of orders increased quite slowly. This can be seen from the slope of two the line

Time

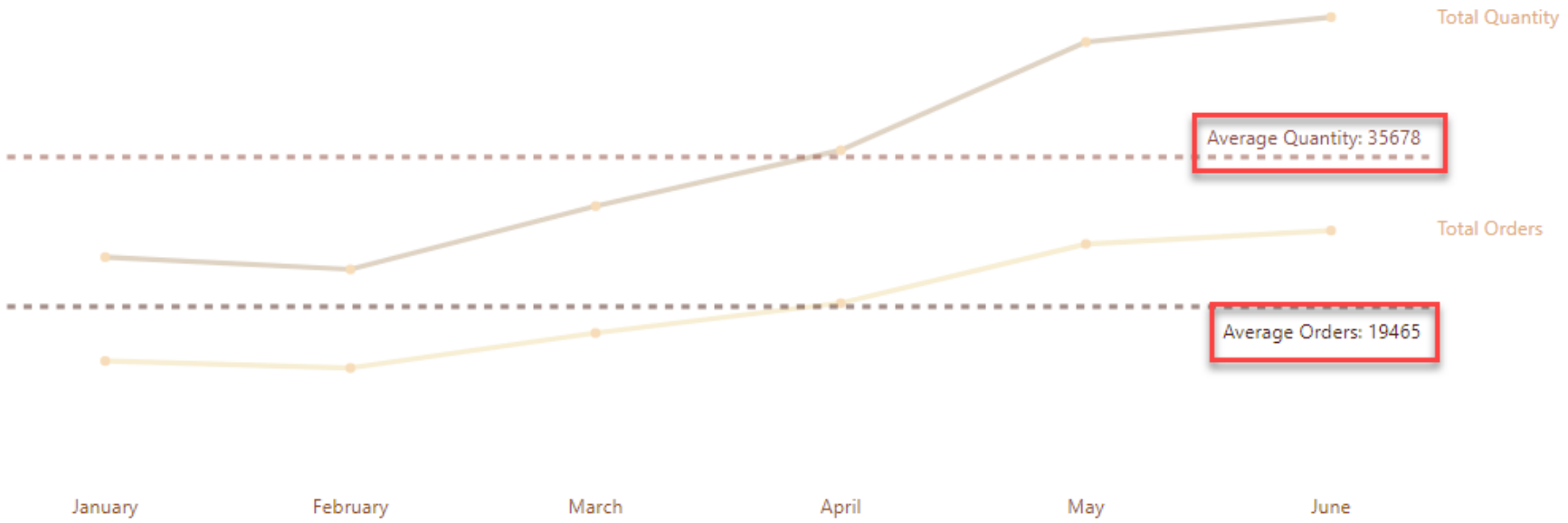
Average Revenue is greater than $116K



- **First 3 months** of the year, total revenue **increased** quite **slowly** and total revenue each month less than average revenue
- Meanwhile, revenue **increased** rapidly in the **second quarter** and was higher than average revenue

Time

Same Total Revenue



- First 3 months of the year, total quantity and total orders increased quite slowly and total revenue each month less than average
- Meanwhile, total quantity and total orders increased rapidly in the second quarter and was higher than average

Time

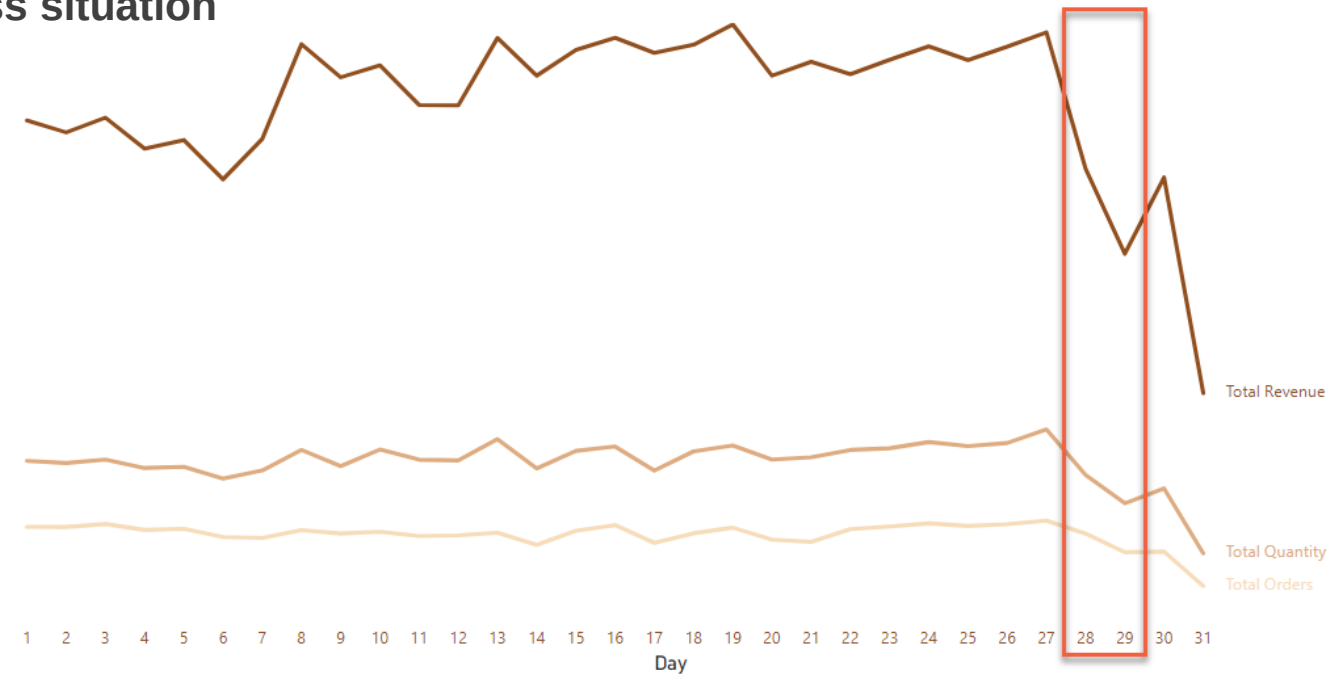
Why do total revenue, total quantity, total orders decrease at February?

category_type	Month Name	AVG Rev	AVG Rev Growth Rate	AVG QTY	AVG QTY Increasing Rate	AVG Orders	AVG Orders Increasing Rate	AVG Sell Price	AVG Sell Price Increasing Rate
Branded	January	90.00	0.00%	4.76	0.00%	4.57	0.00%	18.50	0.00%
	February	72.65	-19.28% ↓	4.35	-8.59% ↓	4.29	-6.07% ↓	16.28	-12.00% ↓
	March	85.76	18.05%	4.95	13.77%	4.67	8.68%	16.97	4.30%
	April	113.29	32.09%	6.76	36.54%	6.48	38.78%	17.03	0.30%
	May	131.32	15.92%	7.32	8.23%	7.27	12.30%	17.85	4.80%
	June	162.52	23.76%	9.29	26.89%	8.76	20.48%	17.48	-2.10% ↓
Complementary Products	January	42.37	0.00%	52.96	0.00%	33.70	0.00%	0.80	0.00%
	February	44.30	4.58%	55.38	4.58%	35.52	5.43%	0.80	0.00%
	March	50.96	15.01%	63.70	15.01%	41.09	15.66%	0.80	0.00%
	April	64.47	26.52%	80.59	26.52%	51.86	26.23%	0.80	0.00%
	May	82.85	28.51%	103.57	28.51%	66.91	29.02%	0.80	0.00%
	June	91.27	10.16%	114.09	10.16%	73.82	10.32%	0.80	0.00%
Finished Products	January	2,326.69	0.00%	746.32	0.00%	436.94	0.00%	3.15	0.00%
	February	2,444.31	5.06%	783.18	4.94%	456.43	4.46%	3.15	0.10%
	March	2,863.75	17.16%	914.71	16.79%	534.77	17.16%	3.16	0.20%
	April	3,542.38	23.70%	1,132.83	23.85%	659.90	23.40%	3.16	0.00%
	May	4,529.88	27.88%	1,449.29	27.93%	844.97	28.04%	3.16	-0.10% ↓
	June	4,933.91	8.92%	1,580.20	9.03%	922.07	9.12%	3.15	-0.10% ↓
Ingredients	January	284.31	0.00%	16.30	0.00%	15.57	0.00%	16.28	0.00%
	February	255.02	-10.30% ↓	15.81	-3.03% ↓	15.71	0.96%	16.41	0.80%
	March	300.14	17.69%	18.65	17.98%	18.61	18.42%	16.51	0.60%
	April	393.36	31.06%	23.05	23.55%	22.23	19.45%	16.20	-1.90% ↓
	May	485.40	23.40%	28.96	25.65%	28.22	26.95%	16.39	1.20%
	June	576.51	18.77%	33.14	14.43%	31.59	11.96%	16.41	0.10%

- All metrics of **Branded** decrease on February
- Average revenue and average quantity increasing rate of **Ingredients** decrease

Time

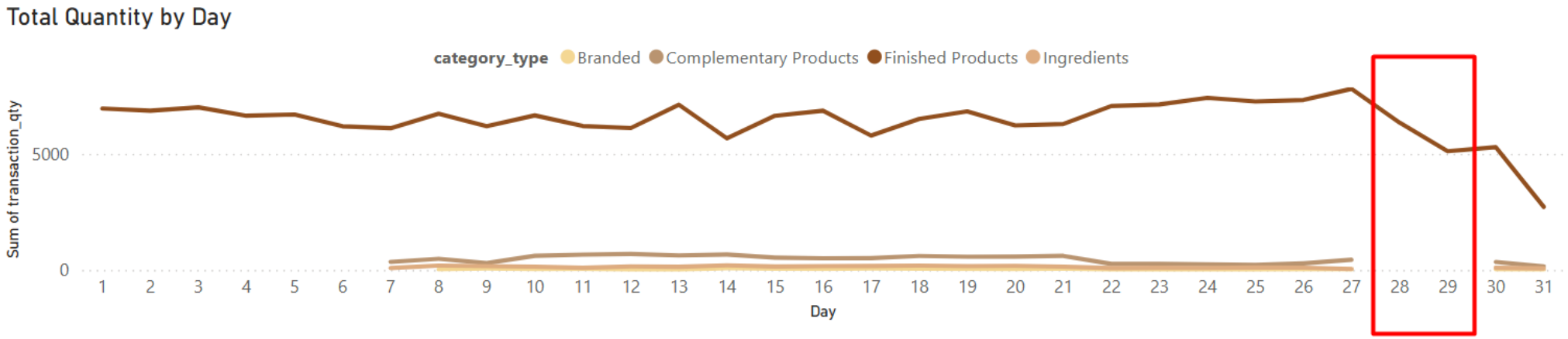
Stable business situation



Total revenue, total quantity and total orders tend to **decrease from 28th to 29th**, then **increase to first month** (~ 6th) and **stable** on **middle of the month**

Time

Why do total revenue, total quantity, total orders decrease from 28th to 29th?



Finished Products are ***bought every day*** of the month. **The other product groups** are usually ***bought from the 7th to the 27th***, then ***again on the 30th and 31st.*** However, their percentage of revenue, sales, and transactions is small, so they do not really affect the drop on the 28th and 29th.

Time

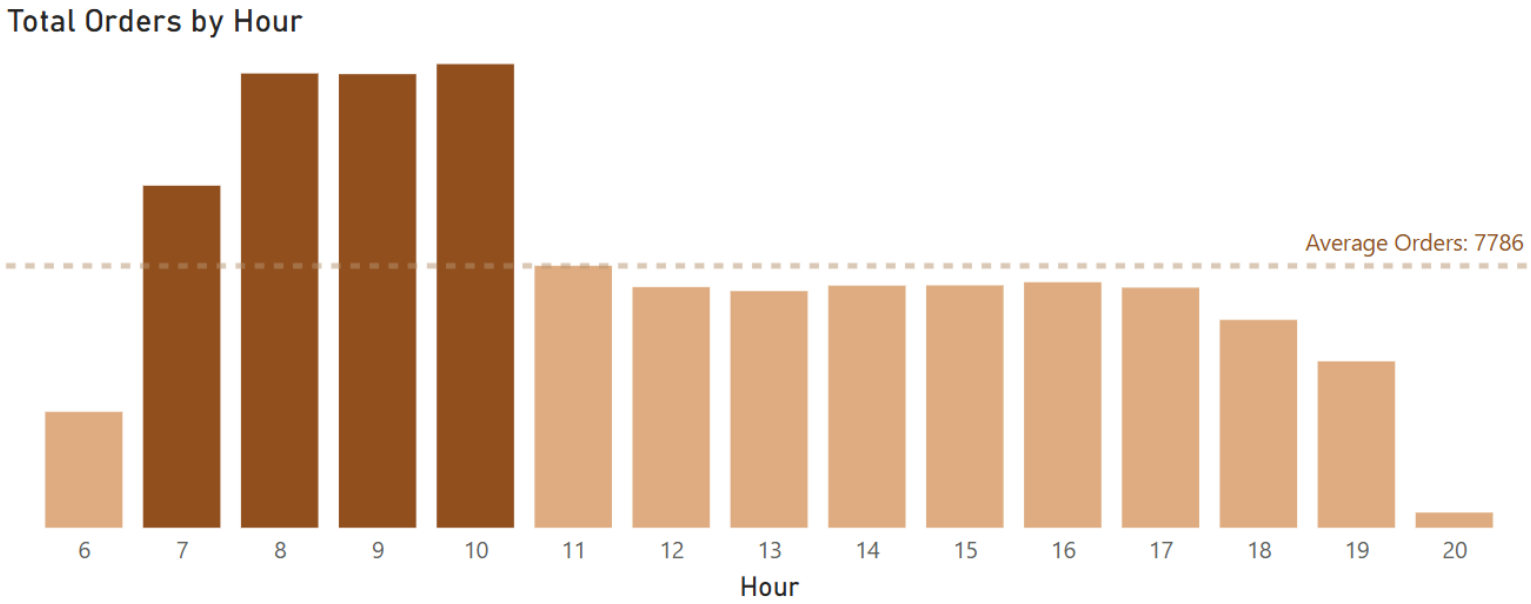
Most categories in Finished Products have lower sales, causing a drop on the 28th to 29th?

	product_category	DAY_28	DAY_29	AVG_2DAYS
1	Bakery	-11.31	13.03	0.8599999999999994
2	Coffee	-10.13	-14.94	-12.535
3	Drinking Chocolate	-29.8	8.54	-10.63
4	Tea	-28.6	-9.28	-18.94

The average sales growth in two days for these categories is negative, except for bakery, which has positive growth but not high. This is the reason for the drop on the 28th and 29th.

Time

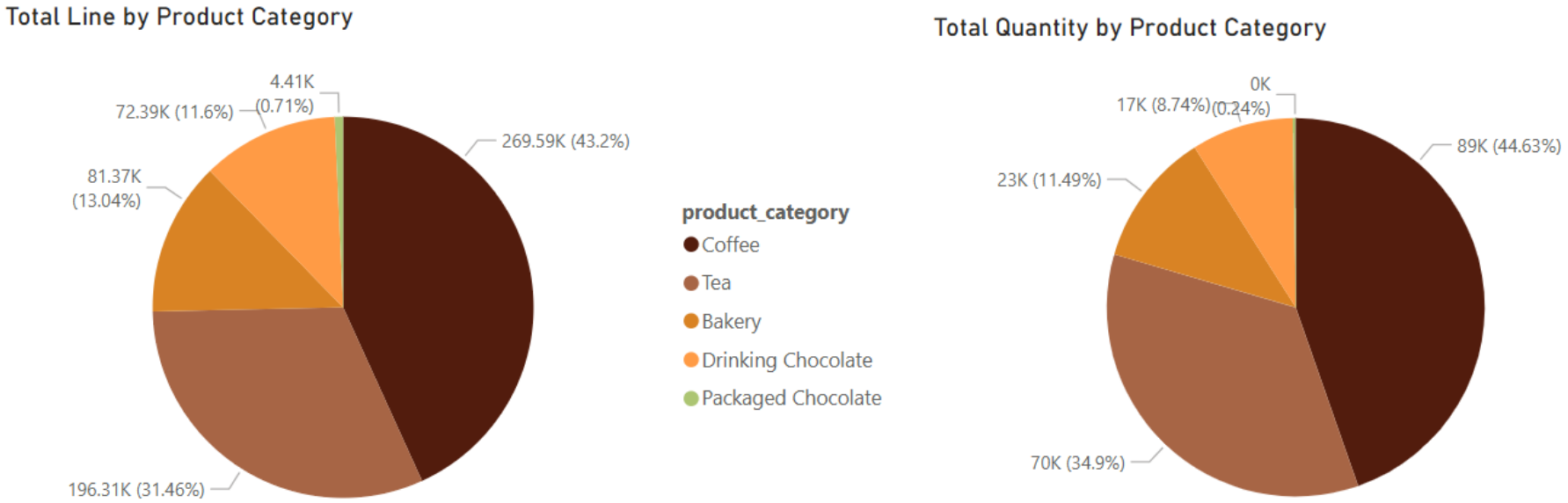
Customers usually buy more in the morning.



Revenue, sales, and orders are higher than average from **7 AM to 10 AM**. Except for 6 AM and 8 PM, which are much lower, other time slots are mostly close to the average.

Products

Finished Products group



Coffee and **Tea** lead in both revenue and quantity, ***highlighting*** their strong market performance.

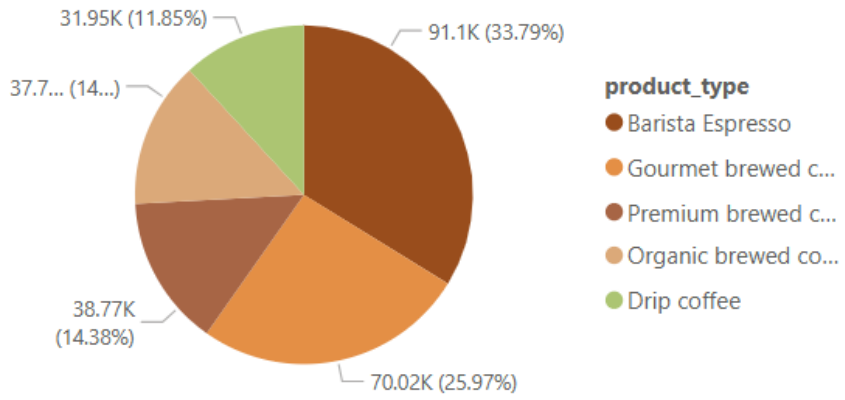
In contrast, **Packaged Chocolate** contributes slightly ***less than 1%,*** likely due to its nature as a *bundled product*.

Products

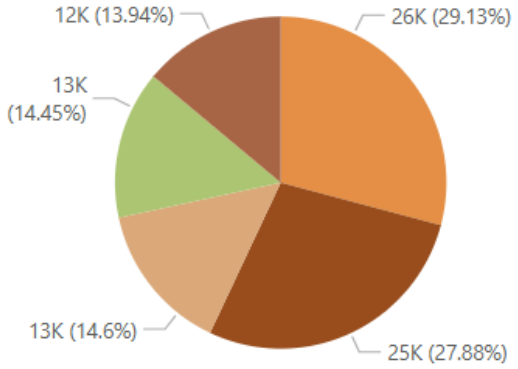
Selling price is a key factor influencing the rankings of most product type within each product category.

Coffee

Total Revenue by Product Type



Total Quantity by Product Type



product_type	Average of unit_price
Barista Espresso	3.66
Premium brewed coffee	3.12
Organic brewed coffee	2.90
Gourmet brewed coffee	2.70
Drip coffee	2.48
Total	**3.02**

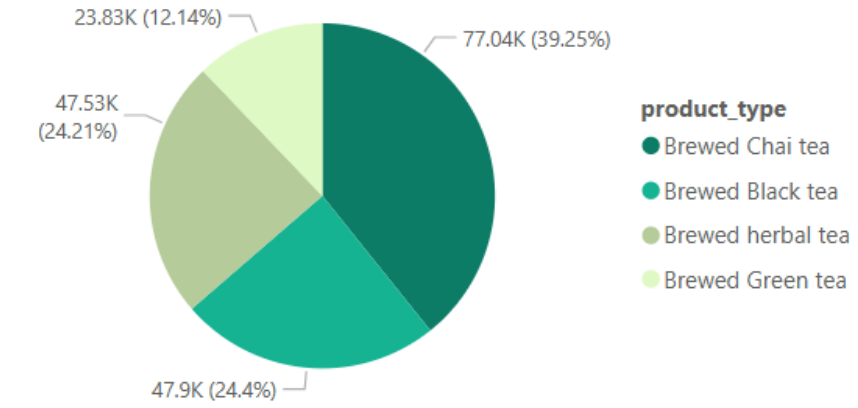
Although **Barista Espresso** and **Gourmet Brewed Coffee** have **similar sales volumes**, **Barista Espresso** **leads in total revenue** due to its notably *higher average unit price*.

Products

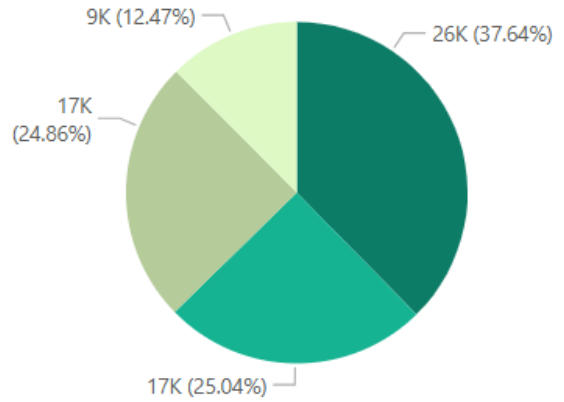
Selling price is a key factor influencing the rankings of most product type within each product category.

Tea

Total Revenue by Product Type



Total Revenue by Product Type



product_type	Average of unit_price
Brewed Chai tea	2.94
Brewed Black tea	2.74
Brewed herbal tea	2.74
Brewed Green tea	2.74
Total	**2.82**

Brewed Chai Tea leads in both **total revenue and quantity**, with **average unit prices** in the category being *relatively consistent*.

Products

Items in the Finished Products category tend to be bought together

product_id	product_detail	product_category
72	Ginger Scone	Bakery
73	Almond Croissant	Bakery
87	Ouro Brasileiro shot	Coffee

product_id	product_detail	product_category
69	Hazelnut Biscotti	Bakery
72	Ginger Scone	Bakery
87	Ouro Brasileiro shot	Coffee

product_id	product_detail	product_category
70	Cranberry Scone	Bakery
72	Ginger Scone	Bakery
87	Ouro Brasileiro shot	Coffee

	product_id	count
31	(72, 73, 87)	24
27	(69, 72, 87)	22
39	(70, 72, 87)	21
37	(72, 77, 87)	19
25	(72, 79, 87)	14
...	...	...
157	(27, 60, 73)	1
158	(46, 60, 69)	1
159	(23, 53, 73)	1
163	(25, 60, 76)	1
200	(22, 47, 79)	1

Customers often **purchase** **coffee** and **bakery** items **together**.

Among these combinations, **Ouro Brasileiro Shot** and **Ginger Scone** are **the most *frequently bought together***.

Products

The unit price of Ginger Scone changes when a second item is purchased.

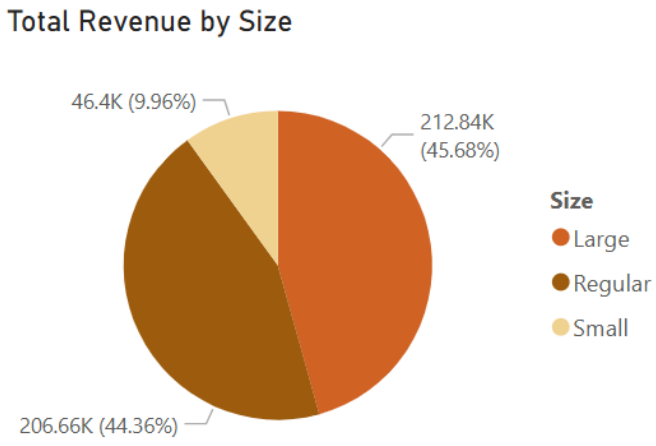
transaction_id	product_id	unit_price	total_qty	total_line	product_category	product_type	product_detail
24481	72	2.65	1	2.65	Bakery	Scone	Ginger Scone
24481	72	3.25	1	3.25	Bakery	Scone	Ginger Scone
24481	87	2.10	1	2.10	Coffee	Barista Espresso	Ouro Brasileiro shot
39343	72	2.65	1	2.65	Bakery	Scone	Ginger Scone
39343	72	3.25	1	3.25	Bakery	Scone	Ginger Scone
39343	87	2.10	1	2.10	Coffee	Barista Espresso	Ouro Brasileiro shot
41972	72	2.65	2	5.30	Bakery	Scone	Ginger Scone
41972	72	3.25	1	3.25	Bakery	Scone	Ginger Scone
41972	87	2.10	2	4.20	Coffee	Barista Espresso	Ouro Brasileiro shot

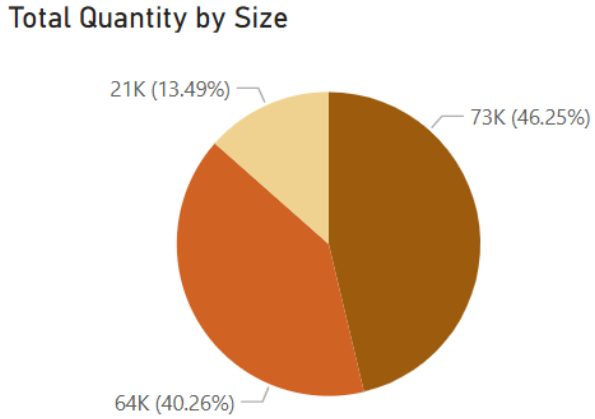
transaction_id	product_id	unit_price	total_qty
63	72	3.25	1
113	72	3.25	1
121	72	3.25	1
172	72	3.25	1
189	72	3.25	1

When **buying a second Ginger Scone**, the ***unit price drops*** from **$3.25 to $2.65**, suggesting a promotional or bundle discount.

Products

Total revenue and quantity sold of Coffee and Tea, categorized by size





product_category	Large	Regular	Small	**Total**
Coffee	3.54	3.09	2.17	**3.02**
Tea	3.14	2.51		**2.82**
Total	**3.33**	**2.81**	**2.17**	**2.93**

Most customers purchase **regular-size** drinks, but since the large size has a higher selling price, **most of the revenue** comes from **large-size** drinks.



Summary

Summary



- Finished Products account for the majority of both total revenue and quantity sold.
- Business performance shows an upward trend.
- Peak customer hours are from 7:00 AM to 10:00 AM.
- Products in the Coffee and Tea categories are frequently purchased.
- The average selling price of Coffee products varies significantly, while prices in other categories are relatively consistent.
- Coffee and Bakery items are often bought together.
- Packaged Chocolate may be an add-on or complementary product.
- Sales typically drop on the 28th and 29th, and some products record zero revenue on these days.



Proposal

Proposal



- Implement discount programs or special events at the end of the month to boost sales.
- Ensure adequate staffing during peak hours.
- Launch promotional offers with free add-ons when customers purchase Coffee and Bakery or Tea and Bakery items.
- Highlight these categories in marketing campaigns and product placement.
- Introduce new seasonal or limited-time coffee/tea items to refresh customer interest.
- Offer morning-only combos (e.g., coffee + bakery) to maximize basket size.
- Launch end-of-month flash deals or time-limited combos to boost engagement.



Thank You